Optimal Siting, Sizing, and Technology Selection of Energy Storage Systems in Distribution Grids

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***Abstract*—The LaTeX template and basic guidelines for the preparation of a technical paper for the PSCC 2026 conference are presented. The abstract is limited to 150 words and cannot contain equations, figures, tables, or references. It should concisely state what was done, how it was done, principal results, and their significance.**

***Index Terms*—The author shall provide up to 5 keywords (in alphabetical order) to help identify the major topics of the paper.**

I. Introduction

The energy transition to low- and zero-carbon technologies has driven vast adoption of distributed energy resources (DERs) such as photovoltaic systems (PVs) and electric vehicles. These technologies are primarily owned by end-users and, therefore, connected to low-voltage distribution networks.

DERs integration presents operational challenges for the distribution system operators (DSOs),i.e., overvoltages and line and transformer congestions. DSOs' compatible methods for these challenges can be mainly classified into two groups: wired and non-wired solutions. A wired solution entails upgrading the physical infrastructure by installing larger-rated cables and transformers. However, current research and market trends have shown non-wired solutions as a more cost-effective method, with a special focus on the utilization of battery energy storage systems (BESS) due to their diverse applications in the grid [1] [2]. On the distribution side of the grid, BESS has shown potential for voltage regulation and congestion management at the transformer side [3].

In this context, in the literature, there are several methods to guide DSOs in computing the optimal energy storage, location, and BESS technology required in the network. These range from heuristics [4], [5], to Optimal Power Flows (OPFs) [6], [7]. In addition, proposed methods differ in how BESS and grid constraints are modelled. Grid constraints are non-linear, non-convex, and not tractable. Therefore, there are several modelling methods proposed in the literature [8] which are shown to be For example, [**?**] Modeling BESS

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II. Methodology

This section proposes a general optimization-based methodology for the optimal siting and sizing of energy storage systems (ESS) to support key distribution grid operations, including voltage regulation, mitigation of line thermal overloads, and transformer-level congestion management.

The proposed framework is designed to accommodate a broad class of ESS technologies by explicitly modeling the operational characteristics that are most relevant for grid interaction. These include energy capacity constraints, converter power limits, charging and discharging efficiencies, C-rate limitations, mutual exclusivity of operating modes, and dynamic constraints such as ramp-rate limits.

By formulating ESS behavior in terms of operational requirements rather than specific technologies, the methodology enables a consistent comparative assessment of heterogeneous storage technologies within a unified grid-constrained optimization problem.

A. Notation

We consider a distribution grid comprising a set of nodes $\mathcal{N} = \{1, \dots, N\}$, a set of distribution lines $\mathcal{L} = \{1, \dots, L\}$, and a set of discrete time steps $\mathcal{T} = \{1, \dots, T\}$. Nodes, lines, and time steps are indexed by $n \in \mathcal{N}$, $l \in \mathcal{L}$, and $t \in \mathcal{T}$, respectively. Furthermore, we consider a set of ESS technologies $\mathcal{J} = \{1, \dots, J\}$, indexed by $j \in \mathcal{J}$.

B. Cost function

We consider three components of the investment cost associated with the siting and sizing of ESS: (i) energy capacity, (ii) converter rating, and (iii) installation cost. The corresponding decision variables are the nominal energy capacity $E_{n,j}$ (kWh), the rated apparent power of the charging-side and discharging-side converters, $S^{\text{conv}+}_{n,j}$ and $S^{\text{conv}-}_{n,j}$ (kVA), and the binary installation variable $I^{\text{inst}}_{n,j} \in \{0, 1\}$, for each node $n \in \mathcal{N}$ and ESS type $j \in \mathcal{J}$.

The energy capacity investment cost is given by

$$\mathcal{F}^{\text{E}} = \sum_{n \in \mathcal{N}} \sum_{j \in \mathcal{J}} c^{\text{E}}_j \, E_{n,j}, \tag{1}$$

where c^{E}_j [CHF/kWh] denotes the unit cost of installed energy capacity.

The converter investment cost accounts separately for the charging-side and discharging-side converters, thus accommodating technologies with asymmetric power-conversion stages (e.g., power-to-X-to-power systems):

$$\mathcal{F}^{\text{conv}} = \sum_{n\in\mathcal{N}} \sum_{j\in\mathcal{J}} \left(c_j^{\text{conv}+}\, S_{n,j}^{\text{conv}+} + c_j^{\text{conv}-}\, S_{n,j}^{\text{conv}-} \right), \quad (2)$$

where $c_j^{\text{conv}+}$ and $c_j^{\text{conv}-}$ [CHF/kVA] are the unit costs of the charging-side and discharging-side converter ratings, respectively.

Finally, we include a lumped installation cost for deploying an ESS of type j at node n:

$$\mathcal{F}^{\text{inst}} = \sum_{n\in\mathcal{N}} \sum_{j\in\mathcal{J}} c_j^{\text{inst}}\, I_{n,j}^{\text{inst}}, \quad (3)$$

where c_j^{inst} [CHF] denotes the fixed installation cost.

C. Net nodal power injections

For each node $n \in \mathcal{N}$ and time step $t \in \mathcal{T}$, the net active and reactive power injections defining the operating point of the distribution grid are obtained from the aggregation of baseline demand, rooftop PV generation, and EV charging demand:

$$P_{n,t}^{\text{net}} = P_{n,t}^{\text{load}} + P_{n,t}^{\text{EV}} - P_{n,t}^{\text{PV}}, \quad (4a)$$

$$Q_{n,t}^{\text{net}} = Q_{n,t}^{\text{load}} + Q_{n,t}^{\text{EV}}. \quad (4b)$$

Here, $P_{n,t}^{\text{load}}$ and $Q_{n,t}^{\text{load}}$ denote the baseline active and reactive power demand at node n and time t, respectively. The terms $P_{n,t}^{\text{EV}}$ and $Q_{n,t}^{\text{EV}}$ represent the active and reactive power consumption associated with EV charging. The quantity $P_{n,t}^{\text{PV}}$ denotes the active power generated by rooftop PV systems, which are assumed to operate at unity power factor and therefore do not contribute to reactive power injection.

Energy storage systems introduce controllable deviations from this operating point through their active and reactive power injections. Since ESS constitute the only flexible resources considered in this work, the deviations in nodal power injections are given by

$$\Delta P_{n,t} = \sum_{j\in\mathcal{J}} P_{n,j,t}^{\text{ess}}, \quad (5a)$$

$$\Delta Q_{n,t} = \sum_{j\in\mathcal{J}} Q_{n,j,t}^{\text{ess}}, \quad (5b)$$

where $P_{n,j,t}^{\text{ess}}$ and $Q_{n,j,t}^{\text{ess}}$ are decision variables representing the active and reactive power injections of ESS technology j at node n and time step t. $P_{n,j,t}^{\text{ess}} > 0$ indicates ESS charging.

D. Linearized grid model

To enforce network operational limits within the optimization problem while maintaining computational tractability, a linearized distribution grid model based on sensitivity coefficients is employed. The linearization follows the approach in [8] and expresses the network states as linear sensitivity-based functions of deviations in nodal power injections.

The grid model is linearized around the net nodal power injections $P_{n,t}^{\text{net}}$ and $Q_{n,t}^{\text{net}}$, which define the operating point of the distribution grid at each node $n \in \mathcal{N}$ and time step $t \in \mathcal{T}$. Deviations from this operating point are induced by controllable resources, namely energy storage systems, and are characterized by the nodal injection deviations $\Delta P_{n,t}$ and $\Delta Q_{n,t}$.

Using sensitivity coefficients derived from a base-case power-flow solution, the main network states, nodal voltage magnitudes, line currents, and transformer loading, are expressed as linear functions of these deviations. Specifically, the nodal voltage magnitudes are given by:

$$v_{n,t} = v_{n,t}^0 + \sum_{m\in\mathcal{N}} \left(K_{v,nm,t}^P\, \Delta P_{m,t} + K_{v,nm,t}^Q\, \Delta Q_{m,t} \right), \quad (6)$$

where $v_{n,t}^0$ denotes the voltage magnitude at the operating point and the sensitivity coefficients $K_{v,nm,t}^P$ and $K_{v,nm,t}^Q$ quantify the impact of active and reactive power deviations at node m on the voltage at node n.

Similarly, the current magnitude on line $l \in \mathcal{L}$ is approximated as

$$i_{l,t} = i_{l,t}^0 + \sum_{m\in\mathcal{N}} \left(K_{i,lm,t}^P\, \Delta P_{m,t} + K_{i,lm,t}^Q\, \Delta Q_{m,t} \right), \quad (7)$$

where $i_{l,t}^0$ is the line current at the operating point and the sensitivity coefficients $K_{i,lm,t}^P$ and $K_{i,nm,t}^Q$ quantify the impact of active and reactive power deviations at node m on the current at line l.

Based on these linearized expressions, the following nodal voltage and line current constraints are enforced for all nodes, lines, and time steps:

$$\underline{v} \le v_{n,t} \le \overline{v}, \quad (8a)$$

$$0 \le i_{l,t} \le \bar{i}_l, \quad (8b)$$

$$(8c)$$

where $\underline{v}$, $\overline{v}$ denote the minimum and maximum operational nodal voltage limits, respectively. $\overline{i_l}$ is the ampacity rating of line l.

The complex power flowing through the main transformer at the slack bus is decomposed into active power, P_t^{S}, and reactive power, $Q_t^{\text{S}}$and is computed as follows for each time step t:

$$P_t^{\text{S}} = P_t^{\text{S},0} + \sum_{m\in\mathcal{N}} \left(K_{sP,m,t}^P\, \Delta P_{m,t} + K_{sP,m,t}^Q\, \Delta Q_{m,t} \right), \quad (9a)$$

$$Q_t^{\text{S}} = Q_t^{\text{S},0} + \sum_{m\in\mathcal{N}} \left(K_{sQ,m,t}^P\, \Delta P_{m,t} + K_{sQ,m,t}^Q\, \Delta Q_{m,t} \right). \quad (9b)$$

Here, $P_t^{\text{S},0}$ and $Q_t^{\text{S},0}$ denote the transformer loading at the operating point and the sensitivity coefficients $K_{sP,m,t}^P, K_{sP,m,t}^Q$ and $K_{sQ,m,t}^P, K_{sQ,m,t}^Q$ quantify the impact of active and reactive power deviations at node m on the active and reactive power at the slack bus respectively including the network

losses. To incorporate the transformer apparent-power limits, $\overline{S}$, we use a box approximation as follows:

$$|P_t^{\mathrm{S}}| \le k_P^{\mathrm{tr}} \overline{S}, \qquad |Q_t^{\mathrm{S}}| \le k_Q^{\mathrm{tr}} \overline{S}. \tag{10a}$$

The utilization factors are chosen such that

$$(k_P^{\mathrm{tr}})^2 + (k_Q^{\mathrm{tr}})^2 \le 1, \tag{10b}$$

ensuring that the rectangular region is a conservative inner approximation of the transformer's P–Q capability curve.

This sensitivity-based grid model enables an explicit and tractable representation of how ESS-induced power-injection deviations affect voltages, line loadings, and transformer utilization, while preserving the linearity of the network constraints and maintaining scalability despite the integer variables introduced in the ESS modelling.

E. Energy storage systems (ESS) modelling

ESS are modelled as controllable grid-side resources whose operation is characterised by active and reactive power injections coupled through inter-temporal state-of-energy dynamics. The formulation is designed to be technology-agnostic and applies to a heterogeneous set of storage technologies, including battery-based and conversion-based systems.

From the grid's perspective, all ESS technologies share a common state-of-energy description. Their specific behaviour is captured by technology-specific efficiency parameters, C-rate limits, and operational constraints such as mutually exclusive operating modes and ramp-rate limits. This structure allows different technologies to be represented within a unified optimization framework, without altering the underlying problem formulation. We also note that all physical and economic parameters are set as exogenous inputs.

1) State-of-energy (SOE) dynamics: To model charging and discharging efficiencies, the active power injection of each ESS is decomposed into charging and discharging components:

$$P_{n,j,t}^{\mathrm{ess}} = P_{n,j,t}^{\mathrm{ess},+} - P_{n,j,t}^{\mathrm{ess},-}, \tag{11}$$

with

$$P_{n,j,t}^{\mathrm{ess},+} \in R_+, \qquad P_{n,j,t}^{\mathrm{ess},-} \in R_+.$$

Here, $P_{n,j,t}^{\mathrm{ess},+}$ denotes the charging power absorbed from the grid, whereas $P_{n,j,t}^{\mathrm{ess},-}$ denotes the discharging power delivered to the grid.

Let $\mathrm{SOE}_{n,j,t}$ denote the stored energy of ESS type j, at node n at time t, and let T_s be the sampling time. The SOE evolves according to

$$\mathrm{SOE}_{n,j,t} = \mathrm{SOE}_{n,j,t-1} + T_s \left(\eta_j^{\mathrm{cha}} P_{n,j,t}^{\mathrm{ess},+} - \frac{1}{\eta_j^{\mathrm{dis}}} P_{n,j,t}^{\mathrm{ess},-} \right), \tag{12}$$

where η_j^{cha} and η_j^{dis} are the charging and discharging efficiencies of technology j, respectively. Equation (12) is applied for all $n \in \mathcal{N}$, $j \in \mathcal{J}$, and $t \in \{1, \dots, T\}$.

The SOE is constrained relative to the installed nominal energy capacity $E_{n,j} \in R_+$:

$$\alpha_j E_{n,j} \le \mathrm{SOE}_{n,j,t} \le (1-\alpha_j) E_{n,j}, \tag{13a}$$
$$\beta_j E_{n,j} \le \mathrm{SOE}_{n,j,0} \le (1-\beta_j) E_{n,j}, \tag{13b}$$

where α_j and β_j are design parameters specifying operational safety margins that are technology j specific.

2) Technology-specific operating requirements: Operational requirements that differ across technologies are defined for different subsets of $\mathcal{J}$. In particular, we define:

- $\mathcal{J}^{\mathrm{MU}} \subseteq \mathcal{J}$ as the set of technologies requiring mutually exclusive charging/discharging,
- and $\mathcal{J}^{\mathrm{RR}} \subseteq \mathcal{J}$ as the set of technologies subject to ramp-rate limits.

For technologies in $\mathcal{J}^{\mathrm{MU}}$, simultaneous charging and discharging is not allowed. This is enforced using a binary mode variable $\mu_{n,j,t} \in \{0,1\}$:

$$P_{n,j,t}^{\mathrm{ess},+} \le \overline{P}_j^{\mathrm{ess}} \, \mu_{n,j,t}, \tag{14a}$$
$$P_{n,j,t}^{\mathrm{ess},-} \le \overline{P}_j^{\mathrm{ess}} \, (1-\mu_{n,j,t}), \tag{14b}$$

where $\overline{P}_j^{\mathrm{ess}}$ is a technology-specific upper bound on the active power magnitude.

Technologies in $\mathcal{J}^{\mathrm{RR}}$ are instead characterised by limited dynamic response, which is captured by ramp-rate constraints on the charging and discharging power:

$$-\Delta_j^+ \le P_{n,j,t}^{\mathrm{ess},+} - P_{n,j,t-1}^{\mathrm{ess},+} \le \Delta_j^+, \tag{15a}$$
$$-\Delta_j^- \le P_{n,j,t}^{\mathrm{ess},-} - P_{n,j,t-1}^{\mathrm{ess},-} \le \Delta_j^-, \tag{15b}$$

where Δ_j^+ and Δ_j^- are technology-specific ramping limits on the charging and discharging power trajectories, respectively.

3) Converter rating and C-rate constraints: Each ESS is interfaced with the distribution grid through power-electronic converters whose rated apparent powers are decision variables in the planning problem. Consistent with the investment-cost formulation in Section 2, separate converters are considered for charging and discharging operation, characterized by the rated apparent powers $S_{n,j}^{\mathrm{conv}+} \in R_+$ and $S_{n,j}^{\mathrm{conv}-} \in R_+$, respectively. This representation accommodates ESS configurations with asymmetric conversion stages, such as power-to-X-to-power systems. For technologies equipped with a single bidirectional converter, symmetric ratings are imposed according to

$$S_{n,j}^{\mathrm{conv}+} = S_{n,j}^{\mathrm{conv}-}. \tag{16}$$

To enforce converter operating limits in a linear and computationally tractable manner, the apparent-power capability of each converter is approximated by an axis-aligned rectangular operating region in the active–reactive power plane. Two utilization factors are introduced: an active-power utilization factor $k_P \in [0,1]$ and a reactive-power utilization factor $k_Q \in [0,1]$.

The active power exchanged with the grid during charging and discharging is bounded by fixed fractions of the corresponding converter ratings:

$$P^{\mathrm{ess},+}_{n,j,t} \leq k_P \, S^{\mathrm{conv}+}_{n,j}, \tag{17a}$$

$$P^{\mathrm{ess},-}_{n,j,t} \leq k_P \, S^{\mathrm{conv}-}_{n,j}. \tag{17b}$$

Reactive power provision is independently bounded by a fixed fraction of the converter apparent-power rating:

$$\left|Q^{\mathrm{ess}}_{n,j,t}\right| \leq k_Q \, S^{\mathrm{conv}+}_{n,j}, \tag{18a}$$

$$\left|Q^{\mathrm{ess}}_{n,j,t}\right| \leq k_Q \, S^{\mathrm{conv}-}_{n,j}. \tag{18b}$$

The parameters k_P and k_Q are chosen such that

$$k_P^2 + k_Q^2 \leq 1, \tag{19}$$

ensuring that the resulting rectangular feasible operating region constitutes an inner approximation of the true, circular, apparent-power capability curve of the converter.

In addition, the admissible charging and discharging powers are constrained by the C-rate of each ESS technology, ensuring consistency between converter sizing and installed energy capacity:

$$P^{\mathrm{ess},+}_{n,j,t} \leq C^{\mathrm{rate}}_{j} \, E_{n,j}, \tag{20a}$$

$$P^{\mathrm{ess},-}_{n,j,t} \leq C^{\mathrm{rate}}_{j} \, E_{n,j}. \tag{20b}$$

where C^{rate}_{j} denotes the maximum C-rate of technology j.

4) Installation-sizing constraint: The following constraints impose the link between the installation site and the sizing of the energy capacity and the converter rating:

$$E_{n,j} \leq E^{\max}_{j} I^{\mathrm{inst}}_{n,j}, \tag{21a}$$

$$S^{\mathrm{conv}+}_{n,j} \leq S^{\max}_{j} I^{\mathrm{inst}}_{n,j}, \tag{21b}$$

$$S^{\mathrm{conv}-}_{n,j} \leq S^{\max}_{j} I^{\mathrm{inst}}_{n,j}, \tag{21c}$$

where $E^{\max}_{j}$ and $S^{\max}_{j}$ are design paramaters and denote the maximum possible energy capacity and converter rating, respectively.

F. Formulation of the planning problem

The planning problem is formulated as a mixed-integer linear problem (MILP). The objective function minimizes the investment cost of ESS, comprising the energy capacity cost, power converter cost, and the installation cost. Collecting decision variables in set

$$\mathcal{X} = \{\mathrm{E}_{n,j}, S^{\mathrm{conv}+}_{n,j}, S^{\mathrm{conv}-}_{n,j}, I^{\mathrm{inst}}_{n,j}, P^{\mathrm{ess}+}_{n,j,t}, P^{\mathrm{ess}-}_{n,j,t}, Q^{\mathrm{ess}}_{n,j,t}, \mu_{n,j,t}\}$$

, then the siting, sizing, and technology selection MILP formulation is as follows:

$$\min_{\mathcal{X}} \mathcal{F}^{\mathrm{E}} + \mathcal{F}^{\mathrm{conv}} + \mathcal{F}^{\mathrm{inst}} \tag{22}$$

subject to:

(4), (5), (6), (7), (9), (8), (11), (12), (13), (14), (15), (16), (**??**), (18), (20)

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III. SCENARIO CONSTRUCTION AND TEMPORAL PROFILES

This section describes the construction of scenario-dependent exogenous inputs used to generate the net nodal injections in (4). In particular, we define (i) how scenario targets derived from SFOE-based projections are mapped to the set of analyzed low-voltage (LV) networks, and (ii) how the corresponding time series of baseline demand, PV generation, and EV charging demand are generated at a 15-minute resolution.